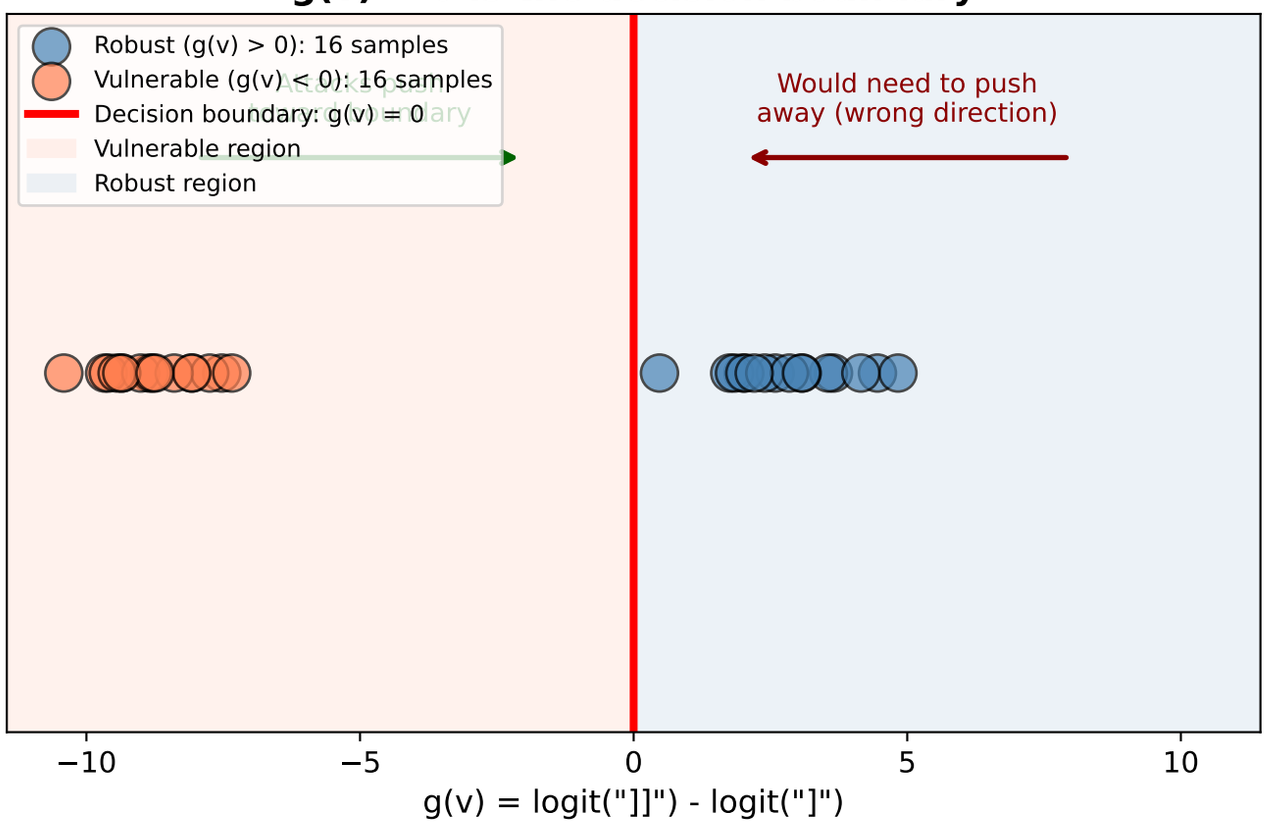
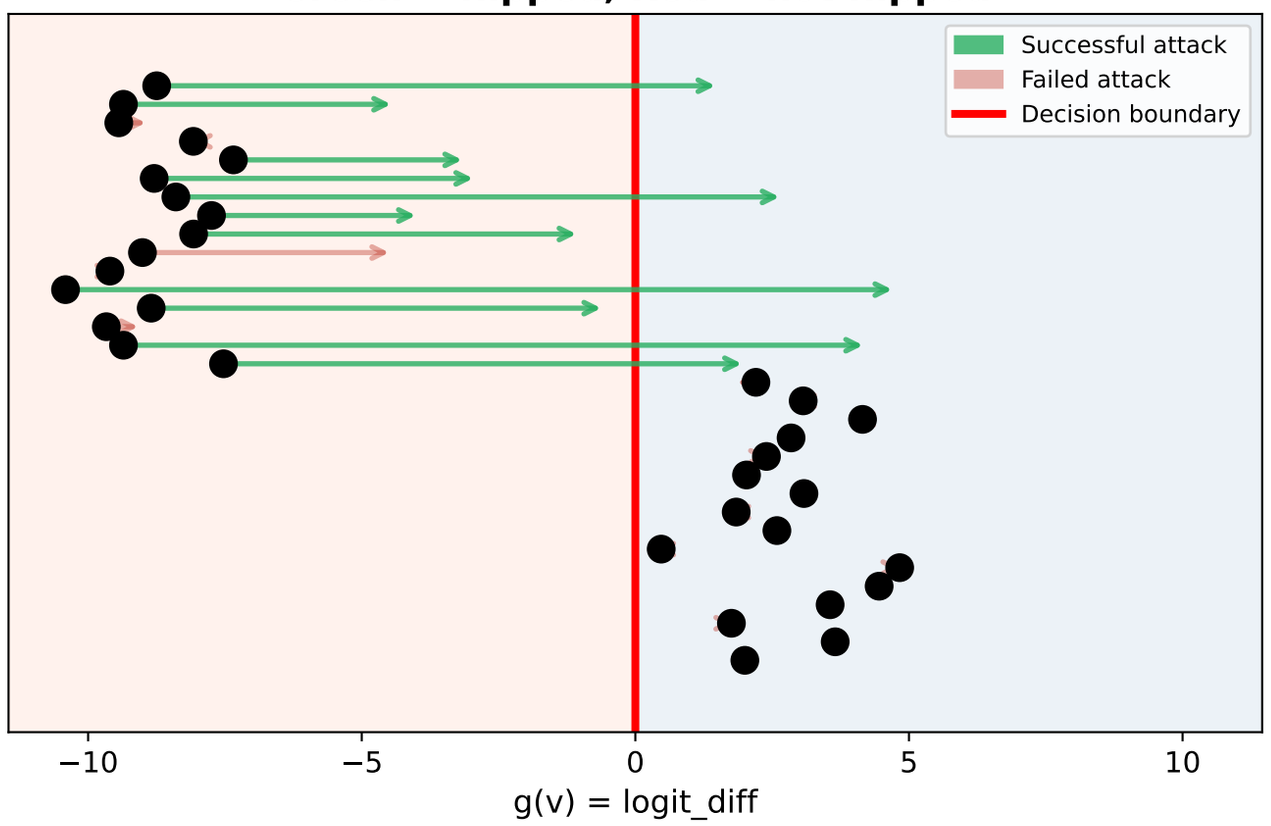
Symmetry Visualization: The Geometry of Adversarial Attacks

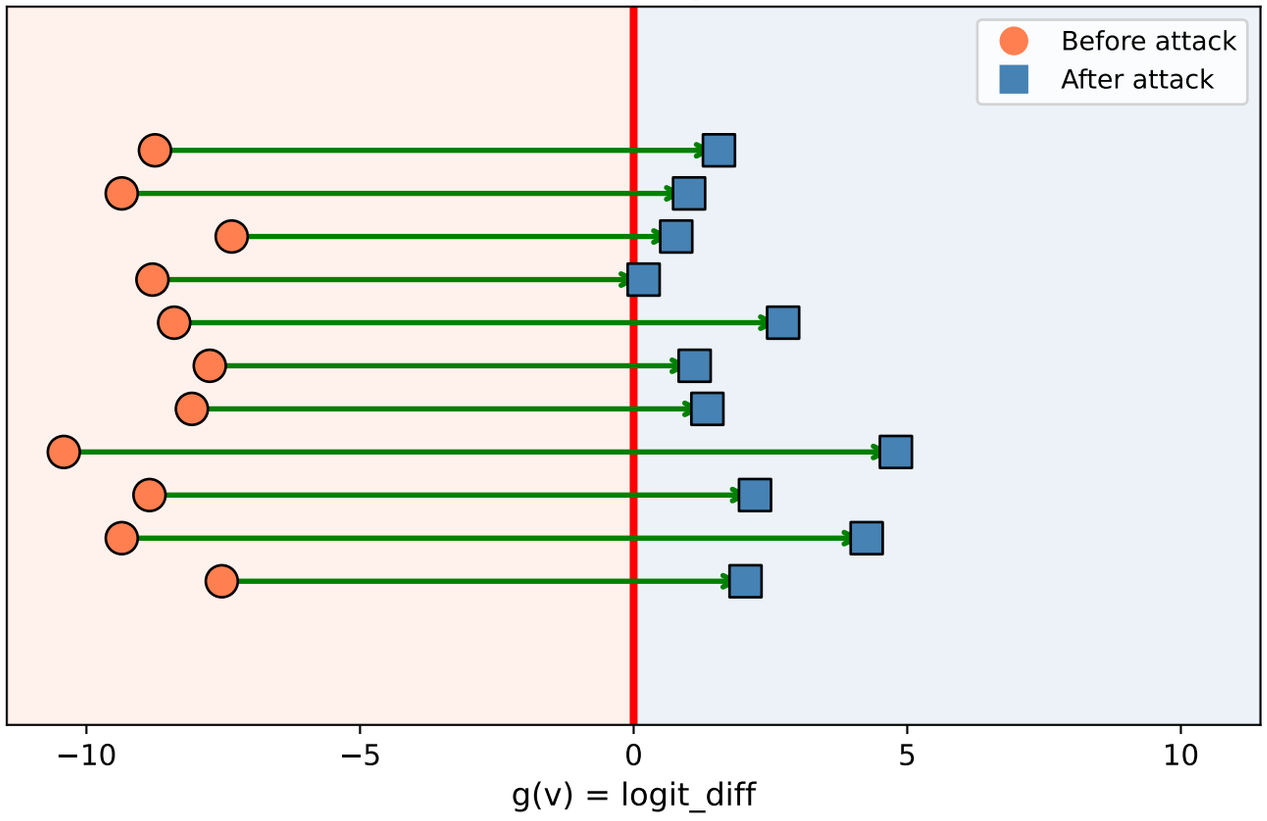
A. Samples on the Symmetry Axis
g(v) = 0 is the decision boundary



B. Attack Vectors (Combined Strategy)
Green = flipped, Red = not flipped



C. Successful Attacks Cross the Boundary
Circle = before, Square = after




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        GEOMETRIC INTERPRETATION: τ-SYMMETRY AXIS
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THE KEY GEOMETRIC INSIGHT
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The Directional Collision Principle has a simple geometric
interpretation:

• g(v) = A(v) - A(τ(v)) = logit_diff
• Decision boundary is where g(v) = 0
• τ-involution swaps the two classes ([] ↔ ])

WHAT THE VISUALIZATION SHOWS
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Panel A: Samples positioned by their g(v) value
         • Vulnerable samples (g < 0) on the left
         • Robust samples (g > 0) on the right

Panel B: Attack vectors showing shift direction
         • Green arrows → successful attacks (crossed boundary)
         • Red arrows → failed attacks (didn't reach boundary)

Panel C: Before/After for flipped samples
         • All successful attacks CROSSED the g(v) = 0 line

WHY DIRECTION MATTERS
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For g(v) < 0 samples (vulnerable):
  → Need POSITIVE shift to reach boundary
  → Circuit attacks achieve this efficiently

For g(v) > 0 samples (robust):
  → Would need NEGATIVE shift
  → Our attacks push in wrong direction → 0% success

This explains the PERFECT SELECTIVITY: attacks only succeed
on samples where they push TOWARD the boundary.

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